

AI-Enhanced Automation Testing with Selenium

Module 1: Introduction to Automation Testing

Core Topics:

- Manual vs Automation Testing
- Benefits and Limitations
- When to Automate

AI Integration:

- Use AI to convert manual test cases into automation-ready scenarios
- Compare Manual vs Automation vs AI-assisted testing

Module 2: Java Basics for Testers

Core Topics:

- Variables, Data Types, Control Structures
- OOP Concepts
- Exception Handling

AI Integration:

- Use AI to generate Java functions
- Auto-complete logic using AI coding assistants

Module 3: Selenium WebDriver

Core Topics:

- WebDriver Setup
- Launching Browsers
- Basic Automation Scripts

AI Integration:

- Generate Selenium scripts using AI prompts
- Understand generated code using AI explanations

Module 4: Locators and WebElements

Core Topics:

- XPath, CSS Selectors
- WebElement interactions

AI Integration:

- Generate XPath/CSS selectors using AI from HTML
- Optimize locators using AI suggestions

Module 5: Handling Web Controls

Core Topics:

- Dropdowns, Checkboxes
- Alerts, Frames, Windows

AI Integration:

- Generate handling code for UI elements using AI
- Practice automation patterns with AI assistance

Module 6: Waits and Synchronization

Core Topics:

- Implicit, Explicit, Fluent Waits
- Handling synchronization issues

AI Integration:

- Debug wait issues using AI
- Explain errors and solutions using AI

Module 7: TestNG Framework

Core Topics:

- Annotations
- Assertions
- Parallel execution

AI Integration:

- Generate TestNG classes and test cases using AI
- Create reports and assertions automatically

Module 8: Framework Design

Core Topics:

- POM, Data-driven, Hybrid frameworks
- Folder structures

AI Integration:

- Design framework architecture using AI
- Generate base classes and reusable components

Module 9: API Testing

Core Topics:

- HTTP Methods
- Postman usage
- Status codes

AI Integration:

- Generate API test cases and payloads using AI
- Create edge cases automatically

Module 10: Selenium Grid

Core Topics:

- Parallel testing
- Cross-browser execution

AI Integration:

- Simplify Grid setup using AI explanations
- Generate configuration scripts

Module 11: Git Integration

Core Topics:

- Version control basics
- Branching and merging

AI Integration:

- Generate commit messages using AI
- Resolve merge conflicts with AI help

Module 12: CI/CD with Jenkins

Core Topics:

- Jenkins setup
- Pipeline creation

AI Integration:

- Generate Jenkins pipelines using AI
- Automate test execution workflows

Final Project: AI-Assisted Automation System

- Generate test cases using AI
- Create automation scripts using AI
- Debug and optimize scripts using AI
- Document where AI failed and human intervention was needed

Tools Covered

- Java
- Selenium WebDriver
- TestNG
- Postman
- Git & GitHub
- Jenkins
- AI Tools (ChatGPT, Copilot, Claude, Codex)